

Nurturing Self-aware Learning through Facial Expression Interpretation

Xingran Ruan*
School of Informatics
The University of Edinburgh
Edinburgh, United Kingdom
xruan@ed.ac.uk

Aurora Constantin
School of Informatics
The University of Edinburgh
Edinburgh, United Kingdom
aconsta2@ed.ac.uk

Charaka Palansuriya
Edinburgh Parallel Computing Centre
The University of Edinburgh
Edinburgh, United Kingdom
charaka@epcc.ed.ac.uk

Kangcheng Wang*
School of Psychology
Shandong Normal University
Jinan, Shandong, China
wangkangcheng@sdsu.edu.cn

Malcolm Atkinson
School of Informatics
The University of Edinburgh
Edinburgh, United Kingdom
malcolm.atkinson@ed.ac.uk

Abstract

A growing focus is on optimizing learning outcomes by supporting metacognitive monitoring, enabling learners to self-assess and adjust their learning strategies. Our goal is to support children in enhancing their metacognitive monitoring skills—the ability to precisely assess and regulate their learning processes. Research shows that targeted strategies can help children improve their skills, provided their imprecise metacognitive monitoring performance (MMP) is identified. However, conventional approaches for identifying imprecise MMP often have low performance and fail to provide the precision needed for effective interventions. To address this, we aim to leverage deep learning to analyze facial expressions as a means of estimating MMP. By interpreting subtle facial expressions, our proposed deep learning approach offers a promising alternative to conventional approaches, enabling more precise and timely estimations of MMP. Our provisional findings suggest significant implications for tailoring strategies for learners' metacognitive monitoring.¹

CCS Concepts

• Human-centered computing → Laboratory experiments.

Keywords

Metacognitive monitoring performance, Children, Deep learning, Facial expression interpretation

ACM Reference Format:

Xingran Ruan, Aurora Constantin, Charaka Palansuriya, Kangcheng Wang, and Malcolm Atkinson. 2025. Nurturing Self-aware Learning through Facial Expression Interpretation. In *Extended Abstracts of the CHI Conference on*

*Corresponding authors.

¹For researchers and developers interested in further exploring or building upon our findings, the neural networks trained in our study will be available at [39].

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720258>

Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3720258>

1 Introduction

Metacognition, the process of reflecting on one's thinking, plays a crucial role in educational success [50]. A key component, metacognitive monitoring, involves evaluating one's learning, performance, and knowledge. The precise value for metacognitive monitoring performance (MMP) represents the case when learners correctly assess their abilities—showing confidence with correct answers and doubt when mistaken [49]. Empirical evidence supports that pupils with higher MMP tend to achieve superior learning outcomes [9, 25, 27]. Recent research underscores that incorporating metacognitive support improves these outcomes [21, 26, 27].

There are pre-defined support strategies to enhance learners' metacognitive monitoring through estimated MMP by conventional approaches using Knowledge Monitoring Assessment (KMA) [21, 26, 27]. Yet, these conventional approaches have limitations in the accuracy of MMP estimation [34, 36]. Stemming from the link between affective states and MMP, we propose a deep learning approach called Meta-Facial Expression Interpreter (M-FEI) for estimating MMP [46, 48]. We present an example of the conventional approach and M-FEI in Figure 1. Unlike traditional AI-based affective computing techniques in education that rely on categorical emotion representations, our M-FEI interprets facial expressions directly. This approach is noted for its objectivity and reliability in recent studies [12].

This study pioneered and developed M-FEI for tailoring metacognitive monitoring support. For instance, if M-FEI identifies imprecise metacognitive monitoring in a learner, it triggers the targeted strategy to provide tailored hints and examples, aiding learners in assessing their knowledge [26, 27]. Conversely, when learners display confidence precisely, the system minimizes guidance, fostering autonomy and maintaining the learning flow. This automatic estimation of MMP maintains learners' flow and engagement by eliminating the need for confidence articulation [3, 29] and enables real-time tailoring of metacognitive monitoring support for individual needs. To clarify, this paper addresses two questions:

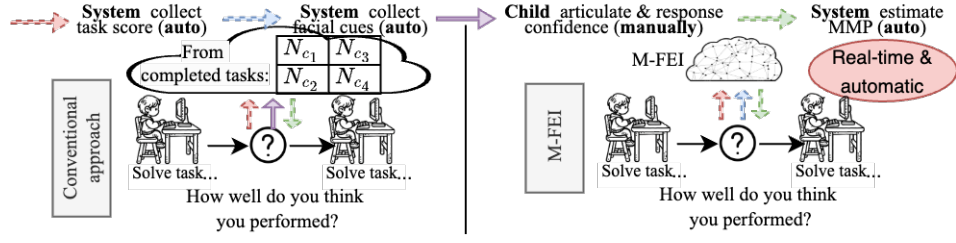


Figure 1: Comparison of the conventional approach and Meta-Facial Expression Interpreter (M-FEI) for estimating MMP. The green arrow of the conventional approach is the estimated MMP based on numbers N_{c1} to N_{c4} . Specifically, N_{c1} is the number of instances that the pupil judged their answer as correct and solved the task correctly; N_{c2} is the number of instances that the pupil judged their answer as correct but solved the task incorrectly. Similarly, N_{c3} counts instances when pupils thought they were wrong yet were correct; and N_{c4} counts cases where pupils thought they were wrong and they were. The green arrow of M-FEI represents MMP estimated based on facial expressions while pupils are reflecting on their solutions. The details are presented in section 4.1 and 4.2.

RQ1 What is the performance of the conventional approach in estimating pupils' MMP?

RQ2 Is it possible to estimate pupils' MMP using deep learning to interpret their facial expressions? If yes, does this improve MMP estimation?

To address these questions, we conducted a user study to develop a dataset of pupil facial expressions, named Affect2Metacognition (A2M), while they engaged in metacognitive monitoring. Then, we developed our M-FEI to estimate pupils' MMP by interpreting facial expressions and compared it with the conventional approach. Based on the comparison, we found that the conventional approach for MMP estimation has limitations in accuracy. In contrast, M-FEI achieved significant enhancements that produced higher precision of imprecise identification with fewer false alarms.

2 Background

2.1 Relationship between Affective States and Metacognitive Monitoring

The complex relationship between learners' affective experiences and metacognitive monitoring during the learning process has been explored in educational research. Metacognitive activities such as evaluating, monitoring, and planning significantly impact learners' affective states. Mandler [33] points out that metacognitive monitoring can trigger visceral arousal and affective responses, especially when learners face cognitive disruptions leading to frustration or confusion [15, 33]. Efklides' Metacognitive and Affective Model of Self-regulated Learning (MASRL) further explores how task characteristics and processing influence both metacognition and affect [17]. Empirical studies by Taub et al. [47, 48] connect specific emotions with metacognitive monitoring, noting that surprise disrupts MMP, whereas frustration may enhance it by motivating learners to tackle challenges.

2.2 Conventional Approach in MMP Estimation

In e-learning platforms, conventional approaches for MMP estimation typically depend on KMA which measures the agreement of task scores with self-rated confidence levels in achieved tasks [36].

For example, Kautzmann et al. use the result of the KMA to adopt a strategy for metacognitive monitoring in the new task. Similarly, research by Guo et al. [21] also employs a variant of the KMA to estimate MMP. However, these conventional approaches have notable limitations. Firstly, the relevance of KMA results diminishes for new tasks that differ significantly in content or domain, making it unreliable for estimating MMP across varied subjects [34, 36]. Additionally, younger learners often lack the metacognitive skills to accurately articulate their confidence, particularly for complex tasks [3, 23, 24, 30].

2.3 Potential of Deep Learning in MMP Estimation

Deep learning has significantly advanced facial expression recognition, particularly for emotions like confusion, frustration, and boredom—common during learning processes. Such technologies are increasingly crucial in educational settings. Singh et al. [43] used deep learning networks [4], achieving a 66.5% accuracy rate, which is a 4.66% improvement over traditional models. Harley et al. [22] found a 60.1% correlation between facial expressions detected by FaceReader and physiological responses measured by electrodermal activity. Meanwhile, Aly [2] demonstrated the effectiveness of pre-trained deep learning networks in identifying learning-related emotional states, providing reliable affect monitoring in educational environments. These findings underscore deep learning's potential to enhance educational technologies by accurately interpreting learner emotions.

To sum up, the link between affective states and metacognitive monitoring, coupled with the feasibility of interpreting facial expressions using artificial intelligence justifies the potential of estimating MMP using deep learning.

3 Data Collection Study

We collected video data while children completing tasks involving cognitive skills—such as inhibitory control, visual and spatial working memory, and target switching—which are fundamental for learning subjects like maths, physics, and others [11, 19, 31]. Our data collection occurred at two primary schools (480 kilometers

		Pupil's Judgement		
		It's correct	I do not know	It's wrong
Pupil's Performance	Correct	$c_1(+ +)$	$c_5(\text{Uncertain})$	$c_3(- +)$
	Incorrect	$c_2(+ -)$		$c_4(- -)$

Figure 2: Five classes of MMP: green represents precise MMP, grey represents uncertain, red represents imprecise MMP.

Table 1: The number of MMP clips in A2M.

Train			Validation			Test		
Imprecise	Precise	Uncertain	Imprecise	Precise	Uncertain	Imprecise	Precise	Uncertain
2084	3543	617	244	386	64	207	440	40

apart) in eastern (Region A) and central (Region B) China. These regions were selected to ensure the generalisability of our findings, accounting for differences in culture, traditions, economic status, urbanization levels, and even physical characteristics such as facial features [42, 51].

3.1 Selection and Participation of Children

The study received ethical approval from the ethics committees of two universities in the UK and China. We initially recruited 185 pupils aged 7 to 11, aligning with Piaget’s stages of cognitive development, where children start to execute logical operations mentally [35]. However, due to technical issues, data from 17 pupils were excluded, leaving a final sample of 168 (80 boys; mean age = 9.83, SD = 1.62). 114 of these were from Region A.

Participation in the study was voluntary, with pupils and their guardians providing informed consent after receiving detailed participant information sheets and an introductory overview. Data confidentiality was ensured by securely storing all materials, including videos and logs, on an encrypted server.

3.2 Procedure of User Study

Our study took place over seven days across two schools, conducting six sessions daily. Each pupil attended one session, scheduled via a preset timetable. In sessions, participants are invited to play Meta-Brainhood (a serious game that develops cognitive skills) [41]. This game has six tasks, each with 10 attempts. Post-attempt, pupils responded to a Judgment of Confidence (JOC) question [49] to assess their confidence in their answers, contributing to measurements of their MMP. The JOC offered three options: ‘No, it is wrong’, ‘Yes, it is correct’, and ‘I do not know’. Five classes of MMP are derived (see Figure 2): $c_1(+ +)$ denotes accurate self-assessment where pupils correctly judge their answers as correct. $c_2(+ -)$ occurs when pupils mistakenly assess incorrect answers as correct. $c_3(- +)$ captures instances where pupils incorrectly deem correct answers as wrong, and $c_4(- -)$ reflects accurate self-judgment of an incorrect answer. Lastly, $c_5(\text{Uncertain})$ applies when pupils are unable to express confidence in their judgment. Our device automatically stores pupils’ clips of addressing JOC and labels clips by derived MMP.

3.3 A2M Dataset from Pupils’ MMP

We built the A2M based on existing strategies to enhance metacognitive monitoring [26, 27], which classifies MMP into imprecise (c_2 or c_3), precise (c_1 or c_4), or uncertain (c_5). For the training and validation phases, we randomly picked 152 pupils’ clips (out of the total 168). Their video clips were randomly allocated, with 90% assigned to the training set and 10% to the validation set. The remaining 16 pupils’ clips formed the test dataset (for simulating the M-FEI’s deployment scenario). This partitioning strategy ensures effective training, validation, and testing of the model by providing distinct datasets that reduce overlap, aligning with best practices in neural network training [32]. The distribution of clips across MMP categories is visualized in Table 1.

4 Methods

4.1 Conventional Approach using the KMA

In this section, we describe the conventional approach using the KMA, which classifies MMP as either imprecise or precise [26, 27]. The conventional approach uses four counts (N_{c_1} , N_{c_2} , N_{c_3} , and N_{c_4}) that capture the occurrences of each MMP category from previous attempts, as depicted in Figure 1. By comparing the KMA index² with threshold θ the MMP is classified as imprecise or precise. We refer to the conventional approach as CA – k , representing the conventional approach using MMP data from the previous k attempts.

4.2 Deep learning Approach: Meta-Facial Expression Interpreter

4.2.1 Feature Extraction. We employed two distinct feature sets to analyze pupils’ facial expressions in the study: the cropped facial region and facial biometric markers. The former captures spatial details of expressions, providing frame-by-frame pixel data ideal for image processing via neural networks. The latter set includes action units (AUs) which have been demonstrated to correlate strongly with MMP in learners³ [14, 16, 37].

²KMA index = $1 - \frac{(N_{c_1} + N_{c_4}) - (N_{c_2} + N_{c_3})}{N_{c_1} + N_{c_2} + N_{c_3} + N_{c_4}}$

³Feature extraction was performed using the OpenFace toolkit [7], known for its robustness [1].

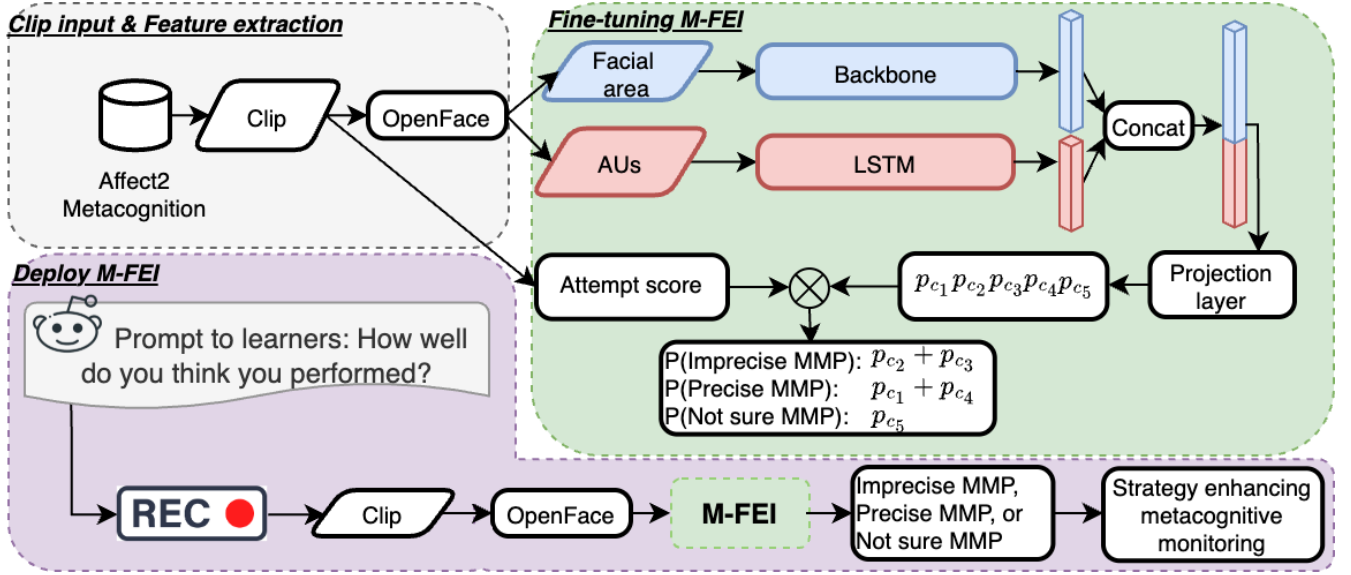


Figure 3: The M-FEI system follows a structured approach to process input clips. Initially, using the OpenFace toolkit [7], we extract two key features from the primary input clips: cropped facial regions and numerical facial AUs, capturing temporal dynamics within each clip. These cropped facial areas are then analyzed through a pre-trained neural network to interpret facial expressions. Simultaneously, the facial AUs are processed by a Long Short-Term Memory (LSTM) network. The system then computes the estimated probabilities, p_{c1} to p_{c5} . Calibration of M-FEI's output is based on the pupil's actual performance, adjusting the probabilities accordingly: if the pupil's answer is correct, p_{c2} and p_{c4} are set to zero; conversely, if the answer is incorrect, p_{c1} and p_{c3} are zeroed.

4.2.2 M-FEI Initialization. We fine-tuned the M-FEI to estimate pupils' MMP, see the green part of Figure 3. After evaluating several state-of-the-art models, we selected the TimeSformer [10] as M-FEI's backbone. For future deployments of M-FEI (purple part), the MMP estimation begins by prompting learners about their perceived performance. Based on the extracted features, the M-FEI performs an inference to estimate the learner's MMP. Further metacognitive monitoring support for pupils' MMP can be tailored based on the identified MMP.

5 Experiments and Results

5.1 Metrics

To critically evaluate CA-k's efficacy, we employed the Area Under the Curve (AUC) to gauge overall performance (score 1→perfect discrimination) [28]. As CA-k is limited in identifying imprecise MMP, our comparison of the M-FEI and CA-k was focused on imprecise, which involved metrics such as True Positive Rate (TPR or recall), False Positive Rate (FPR), and precision [44]. TPR indicates how well the model identifies actual positive instances; FPR quantifies the erroneous positive predictions among negatives; precision reflects the accuracy of these positive predictions [44].

5.2 The Performance of Conventional Approach

To address RQ1, we evaluated the AUC of CA-k. As revealed in Table 2, the AUC progressively increases with the number of attempts, peaking when eight previous attempts are used in both validation and test datasets. The AUC reaches 0.69 with four previous attempts

before plateauing, indicating no further significant gains thereafter.

5.3 M-FEI versus the Conventional Approach

To address RQ2, we first evaluated the feasibility of M-FEI to distinguish between precise, imprecise, and uncertain MMPs. Subsequently, we compared M-FEI's classification efficacy against CA-8, which previously exhibited the highest AUC. Detailed metric outcomes are documented in Table 3.

5.3.1 M-FEI's Performance in Identifying MMP. Table 3 shows M-FEI can identify precise, imprecise, and uncertain MMP, achieving over 70% accuracy across all categories. Notably, M-FEI shows high precisions and low false alarm rates in identifying precise and imprecise MMP. It struggles with uncertain MMP, exposing the challenges of discerning this ambiguous category.

5.3.2 Comparing M-FEI with CA-8. By comparing M-FEI with CA-8, it is evident that M-FEI outperforms CA-8. Particularly, CA-8 consistently underestimated pupils' MMP and overestimated pupils' lack of MMP. While M-FEI achieved higher accuracy (15-17% higher) and exhibited higher precision with a lower number of false alarms (FPR). Taken together, results show that M-FEI is better at detecting pupils who may require extra support (uncertain), while also generating fewer false alarms.

5.3.3 Model Bias: Cross-region Validation. Facial expressions show distinct traits in different locales. Our study was conducted in two

Table 2: AUC values of the CA-k of identifying imprecise in A2M validation and test datasets.

	CA-1	CA-2	CA-3	CA-4	CA-5	CA-6	CA-7	CA-8	CA-9
Validation	0.62	0.65	0.67	0.68	0.69	0.69	0.69	0.69	0.69
Test	0.62	0.63	0.66	0.68	0.68	0.68	0.68	0.69	0.68

Table 3: Comparison of the performance of M-FEI and CA-8 for estimating pupils' MMP

Method	Classify target	A2M							
		Validation				Test			
		Acc.	TPR (Recall)	FPR	Precision	Acc.	TPR (Recall)	FPR	Precision
M-FEI	Precise	0.76	0.80	0.28	0.78	0.71	0.75	0.36	0.79
	Imprecise	0.78	0.70	0.18	0.68	0.74	0.58	0.19	0.56
	Uncertain	0.81	0.58	0.17	0.26	0.74	0.35	0.24	0.08
		A2M (imprecise)							
		Validation				Test			
		Acc.	TPR (Recall)	FPR	Precision	Acc.	TPR (Recall)	FPR	Precision
M-FEI	Imprecise	0.79	0.71	0.16	0.74	0.75	0.58	0.16	0.63
CA-8	Imprecise	0.62	0.77	0.47	0.51	0.60	0.70	0.44	0.43
		Cross-region validation in A2M (imprecise)							
		Region B				Region A			
		Acc.	TPR (Recall)	FPR	Precision	Acc.	TPR (Recall)	FPR	Precision
M-FEI	Imprecise	0.77	0.70	0.21	0.53	0.67	0.63	0.30	0.60
CA-8	Imprecise	0.65	0.69	0.36	0.38	0.60	0.82	0.56	0.52

different regions in China, known for unique physical features, including facial features [38]. We employed a cross-region validation approach for M-FEI, training the model with data from Region A and testing it on Region B, and vice versa. The results, as shown in Table 3, indicate that M-FEI achieved superior accuracy, precision, and a lower FPR in both scenarios, compared to CA-8. These findings underscore M-FEI's reliability and improved generalizability in assessing MMP across diverse regional populations.

5.4 Facial Features that Learned by the M-FEI

Our results show M-FEI outperforms conventional approaches in identifying MMP, thus we employed the Integrated Gradients [45] to calculate attribute values of facial expressions, which illustrate the importance of facial features for MMP identification (1→highest; 0→lowest).

To ensure privacy, we employed a generic face template that aligned with data in our study, see Figure 4 (upper left). The heatmap in Figure 4 (upper right) represents the standardized attribute values, pinpointing crucial facial areas like the brow, eye, nose, cheek, and jaw (in light color). Additionally, the bar graph in Figure 4 (below) displays the attribute values of AUs, with AU04, AU07, AU14, AU17, and AU26 being particularly significant in estimating MMP. These overlaps between the two heatmaps substantially illustrate important areas signaling MMP.

6 Discussion

In this study, we explored the potential of using facial expressions to estimate learners' MMP. We conducted a large-scale user study,

fine-tuning the M-FEI for MMP estimation, and compared its effectiveness against the conventional approach. Our results revealed the potential of M-FEI in interpreting facial cues to estimate MMP. These findings should influence the future development of interventions to enhance metacognitive monitoring.

6.1 Evaluations of Conventional Approach for MMP Estimation

Our study confirms the conventional approach's reliability in estimating MMP [26], yet highlights significant limitations, leading to the advantages of the M-FEI. The conventional approach relies on historical data which could misalign with current tasks. In our study, the performance of the conventional approach reaches the plateau after the fifth attempt. Additionally, our findings underscore issues that pupils are challenged in articulating confidence, with nearly 10% of responses in our dataset marked as uncertain. Moreover, the conventional approach frequently raised false alarms of imprecise, resulting in resource wasting by unnecessary interventions.

6.2 Feasibility of Estimating MMP by Interpreting Facial Expressions

Research on the link between facial expressions and MMP is emerging. To our knowledge, our analysis is the first evidence from a user study supporting the hypothesis that facial expressions reflect pupils' MMP. In our evaluation, we found that M-FEI is feasible in identifying MMP across test and cross-region validation sessions, which demonstrates the feasibility of using M-FEI to estimate MMP from facial expressions. Our work provides a non-verbal, automatic

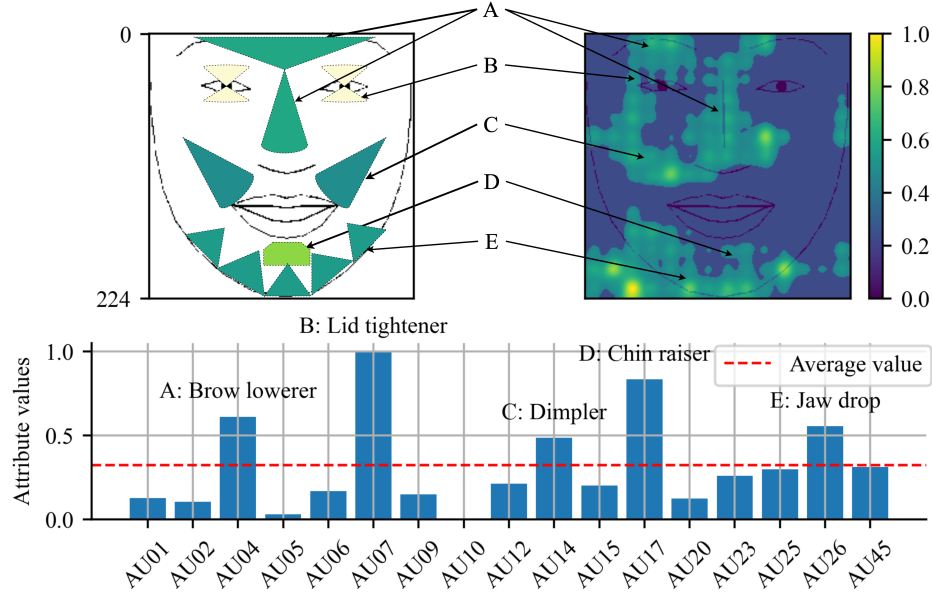


Figure 4: Feature attribute values of M-FEI. Upper left is the template face which is the same as the face area detected by OpenFace in our study; Upper right and lower are the normalized feature attribute values of facial frame pixels and AUs of A2M.

approach to responding to user needs, thereby improving intuitive interaction and accessibility [6, 8].

6.3 Facial Areas Indicating MMP

We conducted experiments to identify the facial areas most informative for estimating MMP. This study pioneers the confirmation that specific facial expressions, including brow lowering, lid tightening, dimple formation, chin raising, and jaw dropping, serve as strong indicators of MMP, with lid tightening being particularly significant. These findings are consistent with prior research on affective states, which links expressions of surprise, confusion, and frustration to MMP [13, 46–48]. Notably, movements in the brows, lids, mouth, and chin are integral to these expressions [18]. Our results reinforce the understanding that facial expressions in these areas are reliable indicators of MMP.

6.4 Application of M-FEI in Educational Setting

Our proposed M-FEI model has the potential to alleviate the workload of educators in large-scale educational contexts, such as massive open online courses (MOOCs), where individual monitoring of each student’s progress is infeasible [5, 20]. By analyzing learners’ facial expressions during self-reflection, the M-FEI could trigger targeted support and raise timely attention to instructors about the potential for imprecise in students’ MMPs. This could stimulate deliveries of targeted interventions to students at risk of being overlooked.

Furthermore, the M-FEI system is engineered with a strong emphasis on ethical standards and data privacy. Unlike systems that

rely on Multi-modal Large Language Models (MLLM), which require advanced hardware, M-FEI is designed to operate locally on the learner’s device⁴. This approach ensures that facial data is collected and processed locally on the device, mitigating risks associated with data breaches and unauthorized access [40, 52]. This local processing model not only enhances data security but also promotes student autonomy by allowing learners to maintain complete control over their personal information.

7 Limitations and Future Work

Our study faces three primary limitations that will be addressed in our future work: Firstly, our study relied on self-reported confidence ratings to label training data, which are susceptible to social desirability and other biases. To address this, we plan to have expert teachers validate JOC responses in A2M dataset. Secondly, while this study includes cross-region validation, we have not yet fully explored how age, ethnicity, or the emotional expressions of individuals from different or better diverse regions affect the performance of M-FEI. An additional user study is underway in the UK, with preliminary results showing promise. Thirdly, we have not yet demonstrated the impact on pupils’ learning outcomes associated with the M-FEI. We are currently running a study to assess both the short-term and long-term educational benefits of M-FEI. The initial findings are positive. They will be published when the studies are completed.

⁴The Total number of parameters in M-FEI is 120 million.

8 Conclusion

Our study assessed MMP estimation using two methods, revealing significant opportunities to adopt interventions for enhancing metacognitive monitoring by interpreting facial expressions. This work introduces three contributions:

1. We revealed the performance and limitations of the conventional approach to estimating MMP.
2. We provided the first evidence from a user study that facial expressions reflect pupils' MMP.
3. We prototyped the M-FEI, which could be deployed for future studies to enhance metacognitive monitoring.

Given the potential of our approach to help learners and teachers, we invite collaborators to help us extend the diversity and depth of research.

Acknowledgments

We gratefully acknowledge the contributions of Xie Wenjing for her assistance in conducting this study. Additionally, we extend our appreciation to the Edinburgh Parallel Computing Center and the Great Britain-China Educational Trust for their support of Xingran Ruan's PhD research.

References

- [1] Sankaraleengam Alagapan, Ki Sueng Choi, Stephen Heisig, Patricio Riva-Posse, Andrea Crowell, Vineet Tiruvadi, Mosadoluwa Obatusin, Ashan Veerakumar, Allison C Waters, Robert E Gross, et al. 2023. Cingulate dynamics track depression recovery with deep brain stimulation. *Nature* 622, 7981 (2023), 130–138.
- [2] Mohammed Aly. 2024. Revolutionizing online education: Advanced facial expression recognition for real-time student progress tracking via deep learning model. *Multimedia Tools and Applications* (2024), 1–40.
- [3] Saleema Amershi, Maya Cakmak, William Bradley Knox, and Todd Kulesza. 2014. Power to the people: The role of humans in interactive machine learning. *AI magazine* 35, 4 (2014), 105–120.
- [4] Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, and Cordelia Schmid. 2021. Vivit: A video vision transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*. 6836–6846.
- [5] Roger Azevedo, Jason Harley, Gregory Trevors, Melissa Duffy, Reza Feyzi-Behnagh, François Bouchet, and Ronald Landis. 2013. Using trace data to examine the complex roles of cognitive, metacognitive, and emotional self-regulatory processes during learning with multi-agent systems. *International handbook of metacognition and learning technologies* (2013), 427–449.
- [6] Serdar Baltaci and Didem Gokcay. 2016. Stress detection in human–computer interaction: Fusion of pupil dilation and facial temperature features. *International Journal of Human–Computer Interaction* 32, 12 (2016), 956–966.
- [7] Tadas Baltrušaitis, Amir Zadeh, Yao Chong Lim, and Louis-Philippe Morency. 2018. Openface 2.0: Facial behavior analysis toolkit. In *2018 13th IEEE international conference on automatic face & gesture recognition (FG 2018)*. IEEE, 59–66.
- [8] Ardhendu Behera, Peter Matthew, Alexander Keidel, Peter Vangorp, Hui Fang, and Susan Canning. 2020. Associating facial expressions and upper-body gestures with learning tasks for enhancing intelligent tutoring systems. *International Journal of Artificial Intelligence in Education* 30 (2020), 236–270.
- [9] Elien Bellon, Wim Fias, and Bert De Smedt. 2020. Metacognition across domains: Is the association between arithmetic and metacognitive monitoring domain-specific? *PLoS One* 15, 3 (2020), e0229932.
- [10] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. 2021. Is space-time attention all you need for video understanding?. In *ICML*, Vol. 2. 4.
- [11] Annie Brookman-Byrne, Denis Mareschal, Andrew K Tolmie, and Iroise Dumontheil. 2018. Inhibitory control and counterintuitive science and maths reasoning in adolescence. *PLoS One* 13, 6 (2018), e0198973.
- [12] Jin Hyun Cheong, Eshin Jolly, Tiankang Xie, Sophie Byrne, Matthew Kenney, and Luke J Chang. 2023. Py-feat: Python facial expression analysis toolbox. *Affective Science* (2023), 1–16.
- [13] Elizabeth B Cloude, Franz Wortha, Daryn A Dever, and Roger Azevedo. 2020. How do Emotions Change during Learning with an Intelligent Tutoring System? Metacognitive Monitoring and Performance with MetaTutor.. In *CogSci*.
- [14] Sidney K D'Mello, Scotty D Craig, and Art C Graesser. 2009. Multimethod assessment of affective experience and expression during deep learning. *International Journal of Learning Technology* 4, 3–4 (2009), 165–187.
- [15] Sidney D'Mello and Art Graesser. 2012. Dynamics of affective states during complex learning. *Learning and Instruction* 22, 2 (2012), 145–157.
- [16] Maria K Eckstein, Belén Guerra-Carrillo, Alison T Miller Singley, and Silvia A Bunge. 2017. Beyond eye gaze: What else can eyetracking reveal about cognition and cognitive development? *Developmental cognitive neuroscience* 25 (2017), 69–91.
- [17] Anastasia Efklides. 2011. Interactions of metacognition with motivation and affect in self-regulated learning: The MASRL model. *Educational psychologist* 46, 1 (2011), 6–25.
- [18] Paul Ekman and Wallace V Friesen. 1978. Facial action coding system. *Environmental Psychology & Nonverbal Behavior* (1978).
- [19] Rachele Fanari, Carla Meloni, and Davide Massidda. 2019. Visual and spatial working memory abilities predict early math skills: A longitudinal study. *Frontiers in Psychology* 10 (2019), 2460.
- [20] Simon Fauvel, Han Yu, Chunyan Miao, Lizhen Cui, Hengjie Song, Liang Zhang, Xiaoming Li, and Cyril Leung. 2018. Artificial intelligence powered MOOCs: a brief survey. In *2018 IEEE international conference on agents (ICA)*. IEEE, 56–61.
- [21] Wenbin Guo. 2020. *Using metacognitive monitoring feedback to improve student learning in augmented reality environments*. Ph. D. Dissertation. University of Missouri-Columbia.
- [22] Jason M Harley, François Bouchet, M Sazzad Hussain, Roger Azevedo, and Rafael Calvo. 2015. A multi-componential analysis of emotions during complex learning with an intelligent multi-agent system. *Computers in Human Behavior* 48 (2015), 615–625.
- [23] Lois R Harris and Gavin TL Brown. 2013. Opportunities and obstacles to consider when using peer-and self-assessment to improve student learning: Case studies into teachers' implementation. *Teaching and Teacher Education* 36 (2013), 101–111.
- [24] Susan Harter. 2012. Developmental differences in self-representations during childhood. *The construction of the self: Developmental and sociocultural foundations* (2012), 27–71.
- [25] Steve Higgins, Maria Katsipatakis, AB Villanueva-Aguilera, Robbie Coleman, P Henderson, LE Major, R Coe, and Danielle Mason. 2016. The Sutton Trust-Education Endowment Foundation Teaching and Learning Toolkit. (2016).
- [26] Tiago Roberto Kautzmann, Talvany Carlotto, and Patricia A Jaques. 2016. Adaptive training of the metacognitive skill of knowledge monitoring in intelligent tutoring systems. In *Intelligent Tutoring Systems: 13th International Conference, ITS 2016, Zagreb, Croatia, June 7–10, 2016. Proceedings 13*. Springer, 301–306.
- [27] Tiago Roberto Kautzmann and Patricia A Jaques. 2019. Effects of adaptive training on metacognitive knowledge monitoring ability in computer-based learning. *Computers & Education* 129 (2019), 92–105.
- [28] Caitlin Kelleher and Wint Hnin. 2019. Predicting cognitive load in future code puzzles. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [29] Kazunori Komatani and Mikio Nakano. 2020. User impressions of questions to acquire lexical knowledge. In *Proceedings of the 21th Annual Meeting of the Special Interest Group on Discourse and Dialogue*. 147–156.
- [30] Florence Kristin LEHNERT. 2024. MEASURING CHILDREN'S USER EXPERIENCE WITH E-ASSESSMENTS: IMPLICATIONS FOR A BETTER INTERPRETATION OF UX EVALUATION METHODS FOR SCHOOL-AGED CHILDREN. (2024).
- [31] Hongxia Li, Xiaoteng Hua, Yalin Yang, Bijuan Huang, and Jiwei Si. 2020. How does task switching affect arithmetic strategy use in children with low mathematics achievement? Evidence from computational estimation. *European Journal of Psychology of Education* 35 (2020), 225–240.
- [32] Shan Li and Weihong Deng. 2020. Deep facial expression recognition: A survey. *IEEE transactions on affective computing* (2020).
- [33] George Mandler. 1989. Affect and learning: Causes and consequences of emotional interactions. In *Affect and mathematical problem solving: A new perspective*. Springer, 3–19.
- [34] Loredana Mihalca and Christoph Mengelkamp. 2020. Effects of induced levels of prior knowledge on monitoring accuracy and performance when learning from self-regulated problem solving. *Journal of Educational Psychology* 112, 4 (2020), 795.
- [35] J Piaget. 1952. The origins of intelligence in children. *International University* (1952).
- [36] Paul R Pintrich, Christopher A Wolters, and Gail P Baxter. 2000. 2. assessing metacognition and self-regulated learning. (2000).
- [37] Mahdi Pourmirzaei, Gholam Ali Montazer, and Ebrahim Mousavi. 2023. AT-TENDEE: an Affective Tutoring system based on facial Emotion recognition and head pose Estimation to personalize e-learning environment. *Journal of Computers in Education* (2023), 1–28.
- [38] Hui Qiao, Jingze Tan, Shaoqing Wen, Menghan Zhang, Shuhua Xu, and Li Jin. 2024. De novo dissecting the three-dimensional facial morphology of 2379 Han Chinese individuals. *Phenomics* 4, 1 (2024), 1–12.
- [39] Xingran Ruan. 2023. Affect2Metacognition. <https://github.com/XRR422/Affect2Metacognition.git>.

- [40] Xingran Ruan, Charaka Palansuriya, and Aurora Constantin. 2023. Affective Dynamic Based Technique for Facial Emotion Recognition (FER) to Support Intelligent Tutors in Education. In *International Conference on Artificial Intelligence in Education*. Springer, 774–779.
- [41] Xingran Ruan, Charaka Palansuriya, Aurora Constantin, and Konstantinos Tsikas. 2023. Supporting Children's Metacognition with a Facial Emotion Recognition based Intelligent Tutor System. In *Proceedings of the 22nd Annual ACM Interaction Design and Children Conference*. 502–506.
- [42] Souphiyeh Samizadeh. 2022. Characteristics of Asian Faces. *Non-Surgical Rejuvenation of Asian Faces* (2022), 41–58.
- [43] Monisha Singh, Ximi Hoque, Donghuo Zeng, Yanan Wang, Kazushi Ikeda, and Abhinav Dhall. 2023. Do i have your attention: A large scale engagement prediction dataset and baselines. In *Proceedings of the 25th International Conference on Multimodal Interaction*. 174–182.
- [44] Stephen V Stehman. 1997. Selecting and interpreting measures of thematic classification accuracy. *Remote sensing of Environment* 62, 1 (1997), 77–89.
- [45] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. Axiomatic attribution for deep networks. In *International conference on machine learning*. PMLR, 3319–3328.
- [46] Michelle Taub and Roger Azevedo. 2018. Using Sequence Mining to Analyze Metacognitive Monitoring and Scientific Inquiry Based on Levels of Efficiency and Emotions during Game-Based Learning. *Journal of Educational Data Mining* 10, 3 (2018), 1–26.
- [47] Michelle Taub, Roger Azevedo, and Nicholas V Mudrick. 2018. How do different levels of AU4 impact metacognitive monitoring during learning with intelligent tutoring systems?. In *Intelligent Tutoring Systems: 14th International Conference, ITS 2018, Montreal, QC, Canada, June 11–15, 2018, Proceedings 14*. Springer, 223–232.
- [48] Michelle Taub, Roger Azevedo, Ramkumar Rajendran, Elizabeth B Cloude, Gautam Biswas, and Megan J Price. 2021. How are students' emotions related to the accuracy of cognitive and metacognitive processes during learning with an intelligent tutoring system? *Learning and Instruction* 72 (2021), 101200.
- [49] Sigmund Tobias and Howard Everson. 1996. Assessing Metacognitive Knowledge Monitoring. Report No. 96-01. *College Entrance Examination Board* (1996).
- [50] Philip H Winne. 2011. A cognitive and metacognitive analysis of self-regulated learning. *Handbook of self-regulation of learning and performance* (2011), 15–32.
- [51] LI Yonglan, YU Huixin, ZHANG Xinghua, YU Keli, BAO Jinping, and ZHENG Lianbin. 2023. Head and facial features of populations in different geographical regions of China. *Acta Anthropologica Sinica* 42, 06 (2023), 793.
- [52] Cheng Zhang, Cheng Chang, Lei Chen, and Yang Liu. 2018. Online privacy-safe engagement tracking system. In *Proceedings of the 20th ACM International Conference on Multimodal Interaction*. 553–554.

Received 23 January 2025; revised 22 February 2025; accepted 22 February 2025